Answers

VARC

1.**A**	2.**C**	3.**C**	4.**D**	5.**B**	6.**D**	7.**B**	8.**D**
9.**D**	10.**D**	11.**C**	12.**B**	13.**A**	14.**C**	15.**C**	16.**B**
17.**C**	18.**2314**	19.**B**	20.**B**	21.**3214**	22.**4312**	23.**B**	24.**A**

LRDI

25.**A**	26.**C**	27.**C**	28.**40**	29.**30**	30.**D**	31.**C**	32.**A**
33.**D**	34.**D**	35.**D**	36.**12**	37.**27**	38.**D**	39.**B**	40.**D**
41.**A**	42.**6**	43.**6**	44.**D**				

Quant

45.**B**	46.**A**	47.**14**	48.**D**	49.**9**	50.**D**	51.**D**	52.**A**
53.**B**	54.**B**	55.**A**	56.**D**	57.**548**	58.**63**	59.**B**	60.**C**
61.**150**	62.**B**	63.**12**	64.**60**	65.**B**	66.**11**		

Explanations

VARC

1.**A**

Option A: *"There was an attempt to formulate Indian culture as uniform, such formulations being derived from texts that were given priority... A dichotomy in values was maintained, Indian values being described as 'spiritual' and European values as 'materialistic', with little attempt to juxtapose these values with the reality of Indian society..."*

It can be understood from the above lines of the passage that the author did not approve(even criticize) the position where one should develop an oppositional framework to grasp cultural differences. Thus, this will not bear any fruit in getting a more accurate view of one nation's history and culture. This is the correct option.

Option B: Throughout the passage, the author criticized the framework the Englishmen adopted to understand India's culture. Thus, the author will support this view that can help give a more accurate picture of a nation's history and culture; hence, this is not the correct option.

Option C: Reading widely into a country's literature without any selection bias will give a more encompassing view of the nation's history; hence, this is also not the correct option.

Option D: This can be rejected on the same ground as option B.

Thus, the correct option is A.

VIDEO SOLUTION

2. **C**
Option A: Since the author criticized the uniform view adopted by the Englishmen to understand Indian culture, the author will support the argument presented in this option. Thus, this is not the correct option.

Option B: This can be understood from the starting and concluding lines of the second paragraph, and hence is not the correct option.

Option C: "*It was a consolation to the Indian intelligentsia for its perceived inability to counter the technical superiority of the west, a superiority viewed as having enabled Europe to colonize Asia and other parts of the world.*"

Although it can be inferred from the above excerpt that the Indians underestimated their culture and knowledge, it cannot be inferred from this excerpt that they matched the technical understanding of the west.

Thus, this view will not be supported by the author and hence, C is the correct option.

Option D: "*...it was believed that the Indian pattern of life was so concerned with metaphysics and the subtleties of religious belief that little attention was given to **the more tangible aspects**.*"

From the above excerpt, it can be inferred that the author disapproved of the Orientalist ignorance of the Indian view towards the materialistic(tangible) aspects. Thus, the author will agree with the view that the Indian culture acknowledges the material aspects of life.

Thus, the correct option is C.

VIDEO SOLUTION

3. **C**
"*It was a consolation to the Indian intelligentsia for its perceived inability to counter the technical superiority of the west, a superiority viewed as having enabled Europe to colonize Asia and other parts of the world. At the height of anti-colonial nationalism it acted as a salve for having been made a colony of Britain.*"

The author mentioned the reference to being a salve in the last paragraph of the passage(above excerpt). The above excerpt was not regarding colonisers; rather, it refers to the Indian intelligentsia (intellectuals or highly educated people as a group). Thus, it can be inferred that option C is not the correct inference and hence is the correct option.

Throughout the passage, it can be inferred that the Orientalist scholars' understanding of Indian history and culture was selective, uniform, generalized, and biased. They viewed the Indian culture largely through the lenses of limited and selected literature in Sanskrit. Thus, option A can be inferred and is not the correct option.

"*A dichotomy in values was maintained, Indian values being described as 'spiritual' and European values as 'materialistic', with little attempt to juxtapose these values with the reality of Indian society. This theme has been even more firmly endorsed by a section of Indian opinion during the last hundred years.*"

From the above lines, option B can also be inferred.

Thus, the correct option is C.

VIDEO SOLUTION

4. **D**
"*There was an attempt to formulate Indian culture as uniform, such formulations being derived from texts that were given priority. The so-called 'discovery' of India was largely through selected literature in Sanskrit. This interpretation tended to emphasize non-historical aspects of Indian culture, for example, the idea of an unchanging continuity of society and religion over 3,000 years*"

From the above excerpt of the passage, it can be inferred that the Orientalist scholars' method of understanding Indian history and culture was selective, uniform, generalized, and biased. They viewed the Indian culture largely through the lenses of limited and selected literature in Sanskrit.

Thus, we need to select an option which resembles the same approach.

Out of the four options, only option D uses a very limited understanding(of selected American movies) to form a generalized view of a nation.

Thus, option D is the correct option.

VIDEO SOLUTION

5. **B**
The passage starts by stating about the sociologists working in the Chicago school tradition on the causality between social disorganization and crime. Then the author describes the immigration experienced in American cities in the 1920s and 1930s. The author then gives the reason why this led to an increase in crime rates(some examples being *failure to integrate these immigrants, coupled with other forces of social disorganization, such as crowding, poverty, and illness, caused crime rates to climb in the cities, particularly in the segregated wards and neighborhoods where the migrants were forced to live.*)

Both options A and C should be eliminated because, in these options, the words social disorganization/ organization or crimes are missing. Compared with D, B is better because the term heavy industry is not a keyword of the passage. Also, more than population growth, migration is the primary reason behind social disorganization.

Thus, the correct option is B.

VIDEO SOLUTION

6. **D**
The above passage is a case study of how rapid or dramatic social change causes relate to increasing crime growth in Chicago. The passage focus on the effects of social disorganization on crime in Chicago.

Option A: There is no comparison of crime in Chicago with that of crime in other states. Thus, this is not the correct answer.

Option B: The passage focus on the effects of social disorganization on crime in Chicago. It is not specific only to the racial aspect. Thus, this is not the correct option.

Option C: This is a distortion of the passage's main idea. Thus, this is not the correct option.

Option D: This option aptly describes the main conclusion of the passage and hence, is the correct option.

The correct option is D.

VIDEO SOLUTION

7. **B**

"*Failure to integrate these immigrants, coupled with other forces of social disorganization such as crowding, poverty, and illness, caused crime rates to climb in the cities, particularly in the segregated wards and neighborhoods where the migrants were forced to live.*"

From the above excerpt of the penultimate paragraph of the passage, it can be inferred that poverty and the rise in disorganization contributed to the increase in crime in American cities. Thus, option A is a valid inference and hence can be eliminated.

"*The social lives of these migrants, as well as those already living in cities they moved to, were disrupted by the differences between urban and rural life . According to social disorganization theory, until the social ecology of the "new place" can adapt, this rapid change is a criminogenic influence.*"

From the above excerpt of the second passage, it can be inferred that the difference between urban and rural life contributed to the disruption experienced by the migrants. Also, this rapid change contributes to the rise in crime in these cities. Thus, options C and D are also valid and hence cannot be the answer.

"*These migrants, unlike their white counterparts, were not integrated into the cities they now called home. In fact, most American cities at the end of the twentieth century was characterized by high levels of racial residential segregation.*"

Although the African American migrants faced a high level of racial segregation, it is nowhere mentioned that it was because they were less organised. Thus, option B is not true in the scope of the passage and hence is the correct option.

Thus, the correct option is B.

VIDEO SOLUTION

8. **D**

Option A: If the workforce size is the largest in the rural area throughout the twenty-first century, then it directly contradicts the theory of the migration of most of the population from rural areas to urban areas. The sample space of the workforce population does not tally with this theory of migration. Thus, this is not the correct option.

Option B: If this population census of 1952 is accurate, it directly nullifies the above migration theory. Thus, this is not the correct option.

Option C: If the data for the estimation of per capita income in the mid-twentieth century primarily required data from the rural areas, then it is not possible that the majority of the population lives in the urban areas. Thus, this is not the correct option.

Option D: If this option is true, it will strengthen the theory of the intermigration of people from rural to urban areas. Thus, this is the correct option.

Thus, the correct option is D.

VIDEO SOLUTION

9. **D**

The starting two paragraphs discuss the complexity of the biosphere and how it is impossible to build a thinking device without bio-logic. In the next paragraphs, the author describes how with the increasing complexity of human-made systems(not until it was comparable to living things), it has become possible to transfer these traits into mechanical systems. Examples of these are bioengineering and genetic engineering. Then in the concluding paragraph, the author discusses about the convergence of these two logics(Biologic and Techno logic).

Options B and C do not talk about the conclusion of the passage(convergence of the logics), and hence can be eliminated. Out of options A and D, we should select the option with bio-logic and techno-logic instead of carrots and cows, because the broader idea is about bio and techno, not carrots and cows.

Thus, the correct option is D.

VIDEO SOLUTION

10. **D**

*"The overlap of the mechanical and the lifelike increases year by year. **Part of this bionic convergence is a matter of words**. The meanings of "mechanical" and "life" are both stretching until all complicated things can be perceived as machines, and all self-sustaining machines can be perceived as alive."*

From the above line, the author tries to show the increasing similarities between 'mechanical' and 'lifelike' with the passage of time. He states that this increase in similarities will continue till the meanings and the perception of the words become synonymous.

Option A: This option states the opposite of what the author tried to convey and hence is not the correct option.

Option B: This option is distorted and can be rejected on the same grounds as option A.

Option C: This is a distorted inference, and the author did not use the above statement to show the meeting grounds of 'genetic engineering' and 'mechanical engineering'. Thus, this is not the correct option.

Option D: This option aptly expresses the point made by the author in the last paragraph, and hence is the correct option.

Thus, the correct option is D.

VIDEO SOLUTION

11. **C**

"*Yet beyond semantics, two concrete trends are happening: (1)Human-made things are behaving more lifelike, and (2) Life is becoming more engineered. The apparent veil between the organic and the manufactured has crumpled to reveal that the two really are, and have always been, of one being..*"

The main argument made by the author in the last paragraph is regarding the increasing similarities between manufactured and organic(lifelike) reality. According to the author, the growing similarities(because of the scientific advances) have distorted the understanding of the realities and have made us think that perhaps these two are and have always been the same.

Option A: This is a distorted inference. It is not that the Organic reality has crumpled under the veil of manufacturing; instead, their meanings are converging mutually. Thus, this is not the correct option.

Option B: This is again a distorted inference. It is not the organic veil that has crumpled; instead, it is the apparent veil. Similarly, in the second half of the option, the organic reality is replaced with the apparent reality.

Option C: This option aptly expresses the main point of the author and is the correct option.

Option D: The author nowhere stated or implied this, and hence this option can be easily eliminated.

Thus, the correct option is D.

VIDEO SOLUTION

12. **B**
*"Although many philosophers in the past **have suspected** one could abstract the laws of life and apply them elsewhere, it wasn't until the complexity of computers and human-made systems became as complicated as living things that it was possible to prove this."*

Option A can be easily rejected from the above excerpt from the passage. Also, it can be inferred that now(not before), since the complexity of computers and human-made systems are comparable, the logic of Bios can be applied to machines. Although option C seems to convey the same meaning, it generalises the complexity and is a distorted inference.

The author has nowhere mentioned or implied in the passage that purposeful design represents the pinnacle of scientific expertise in the service of human betterment and civilisational progress. Thus, option D can also be rejected.

"Genetic engineering is precisely what cattle breeders do when they select better strains of Holsteins, only bioengineers employ more precise and powerful control. While carrot and milk cow breeders had to rely on diffuse organic evolution, modern genetic engineers can use directed artificial evolution—purposeful design—which greatly accelerates improvements."

From the above excerpt from the penultimate paragraph, it can be inferred that although genetic engineering has less control over the products than bioengineering, they both try to evolve the product artificially. Thus, option B can be inferred from the passage.

Thus, the correct option is B.

VIDEO SOLUTION

13. **A**
"*In a recent paper published in the journal Diagnosis, three medical researchers . . . examined the misdiagnosis of Thomas Eric Duncan, the first person to die of Ebola in the U.S., at Texas Health Presbyterian Hospital Dallas. They argue that the digital templates used by the hospital's clinicians to record patient information probably helped to induce a kind of tunnel vision.*"

From the above expert, we can infer that the misdiagnosis of the ebola patient could have been caused by the digital templates used. The information stored in the templates may have helped induce tunnel vision, so the diagnosis could not capture the virus.

" *"These highly constrained tools", the researchers write, "are optimized for data capture but at the expense of sacrificing their utility for appropriate triage and diagnosis, leading users to miss the forest for the trees". Medical software, they write, is no "replacement for basic history-taking, examination skills, and critical thinking.". . .*"

The subsequent excerpt gives information about such tools, which are primarily used to capture/store data at the expense of appropriate diagnosis, which leads the users to miss the more important information[to miss the forest for the trees].

Thus, the major culprit, in this case, is the less important data captured by the doctors, which led them to think in the wrong direction.

Since only option A puts the onus on the data processed by the digital templates, this is the correct option.

Options B, C, and D can be eliminated on the basis of the explanation provided above.

VIDEO SOLUTION

14. **C**
"*There is an alternative. In "human-centred automation," the talents of people takeprecedence. . . . In this model, software plays an essential but secondary role. It takes over routine functions that a human operator has already mastered, issues alerts when unexpected situations arise, provides fresh information that expands the operator's perspective and counters the biases that often distort human thinking. The technology becomes the expert's partner, not the expert's replacement.*"

The above excerpt from the passage defines and applies the human-centred approach. This model should have humans as the primary mind, and the software's rule should be restricted only to assistance.

Option A: Since the role of the software is only specified to the feedback on the doctor's analysis, this is a perfect example of the hum-centred approach. Thus, this is not the correct option.

Option B: In this option, too, the role of technology is dependent on the instructions provided by the resident(human), and hence, it is not the correct option.

Option C: Since the software, in this case, operates on its own(auto-completion), it does not take account of human talent and thinking and hence, is not an example of human-centred automation. Thus, this is the correct option.

Option D: In this case, the software only works or provides assistance when the user requests, and hence, it is not the correct option.

Thus, the correct option is C.

VIDEO SOLUTION

15. **C**
Option A: From the findings of the information scientists at Utrecht University research(2nd paragraph), it can be concluded that the excessive usage of sophisticated software stunted the thinking and learning of its users. Thus, this is not the correct option.

Option B: From the penultimate paragraph of the passage, it can be inferred that the overemphasis on the data by these 'highly constrained tools' can lead the users astray from their desired target. Thus, this option is also not the correct option.

Option C: Nowhere in the passage is it mentioned or implied that the software can replace human beings. On the contrary, the example of the research paper in the journal Diagnosis points to the limitation of these 'sophisticated' softwares. Thus, this is the correct option.

Option D: This can be inferred from the second and third paragraphs of the passage.

Thus, the correct option is C.

VIDEO SOLUTION

16. **B**

"*The researchers found that the people using the simple software developed better strategies, made fewer mistakes and developed a deeper aptitude for the work. The people using the more advanced software, meanwhile, would often "**aimlessly click around**" when confronted with a tricky problem. The supposedly helpful software actually short-circuited their thinking and learning.*"

The above excerpt gives the findings of the Utrecht University experiment. The two study groups(one assisted by simple software and the other by a more sophisticated one) show contrasting behaviours. When confronted with a tricky problem, the one with the advanced software would often aimlessly click around the screen to solve the problem. This shows the dependent behaviour of the user on the software. In other words, the users, rather than trying to develop a strategy for the problem, were expecting it to get done by the software.

Option A: Nowhere in the excerpt was the competency of the users questioned. Instead, it was the effect of the dependency on the software being tested. Thus, this option cannot be inferred.

Option B: The users expected the software to help with the tricky problems. This was described by the phrase "aimlessly click around" in the above excerpt. Thus, this is the correct option.

Option C: The phrase "aimlessly click around" was not used to contrast the strategies adopted by the two study groups; hence, this is not the correct option.

Option D: This again cannot be inferred from the above excerpt.

Thus, the correct option is B.

VIDEO SOLUTION

17. **C**

The main points of the paragraph are:

i) The copying of fashion ideas unique to particular cultures or heritages is rising in this age of social media.

ii) The original communities are not credited and compensated when their unique ideas are used.

Option A: This is a distorted option. It is generalizing that copying a fashion idea is tantamount to stealing(not specifying whether it is done with or without the consent of the original communities.).Thus, this is not the correct option.

Option B: Again, this is a very general and extreme option. Also, it is a distorted inference that the media has encouraged mass production. Thus, this is also not the correct option.

Option C: Since this includes both the main points, this is the correct option.

Option D: This is a distorted option and does not include the main ideas of the paragraph.

Thus, the correct option is C.

VIDEO SOLUTION

18. **2314**

A brief reading of sentences 1 and 4 tells us that they form a pair. In sentence 1, the author described how he/she was made to feel guilty about not going outside while staying Indoors. Statement 4 describes how it felt like a threat for an introverted kid. Thus, 1-4 is a pair.
Now, out of sentences 2 and 3, 2 initiates the topic of discussion[how the author was impelled by her mother to go in the sun]. Statement 3 gives the information about the tales and folklore given to the author to impel. Thus, 2-3 is a pair.
1 is a direct follow-up to 3; hence, the correct order is 2-3-1-4.

VIDEO SOLUTION

19. **B**

The sentence would best fit Blank 2 because it ties together the ideas presented in the paragraph. The paragraph describes the dual inheritance theory. In the first two lines, the author states the theory. The given sentence extends the idea and gives the implication of the theory. After blank 2, the author exemplifies the implication given in the sentence(problem sentence) in the following line.

Thus, the correct option is B.

VIDEO SOLUTION

20. **B**

The main ideas of the paragraph are:

i) The alphabetical order did not directly follow the discovery of alphabets.

ii) Scholars were reluctant to categorize the alphabet in the middle ages because of the fear of rejection of the divine order.

iii) Only after the rediscovery of Greek and Roman classics and the Government bureaucracy in later centuries did the categorization happen.

Option A misses capturing the point of why scholars were reluctant to categorize things according to the alphabet.
Option C is factually incorrect in mentioning the ban on the use. Option D can be eliminated on the same grounds as option A.

Option B captures the essence of the main points most aptly and hence, is the best answer.

Thus, the correct option is B.

VIDEO SOLUTION

21. **3214**
A brief reading of the sentences suggests that the paragraph is about the limits to considering meritocracy as an organising principle in a free society. Statement 3 introduces the topic at hand by stating the inevitability of meritocracy. Statement 2 initiates the main idea by citing the general idea given in recent books about the limits of merit and its applicability. Thus, 3-2 will form a pair.
Statement 1 extends this idea by giving the benefits of understanding the limits, and statement 4 follows this by applying this to build a more compassionate society.

Thus, the correct sequence will be 3-2-1-4.

VIDEO SOLUTION

22. **4312**
A brief reading of the sentences suggests that the paragraph is about beacons and their applications. Statement 4 introduces the topic by giving the definition and its utility. Statement 3 sheds light on the working of beacons. Thus, statement 3 will follow statement 4.

Sentences 1 and 2 discuss the utility of these beacons in different industrial sectors, with statement 2 explaining how the beacon would help retailers.

Thus, the correct order will be 4-3-1-2.

VIDEO SOLUTION

23. **B**
The sentence would best fit in Blank 2 because it ties together the ideas presented in the paragraph. The paragraph describes the problems in getting back the employees in the office. In the first two lines, the author mentions this problem. The given sentence gives the effect of the problems and hence will occupy the blank 2. Also, after blank 2, the author changes the flow of the idea and starts describing the present scenario regarding the steps taken by the companies to face this issue.

Thus, the correct option is B.

VIDEO SOLUTION

24. **A**
The main ideas of the passage are:

i) The job of a joke is to offend its target(victim) irrespective of its status.

ii) The cancel culture deems it inappropriate to joke about people deemed lower in society.

Option A: This option includes both the main points and hence is the correct answer.

Option B: This is a distorted option. The ideas in the paragraph are not intended to persuade to include people from the lower class in the joke. Thus, this is not the correct answer.

Option C: Again, this is a distorted option and can be eliminated based on the explanation given in option B.

Option D: This is also a distorted option, as nowhere in the passage the duties of a comedian are mentioned. Thus, this is not the correct option.

Thus, the correct option is A.

VIDEO SOLUTION

LRDI

25. **A**
The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males alive in 2000 = 180 + 205 + 160 + 100 = 645

Thus, the number of dead males in 2000 = 1000 - 645 = 355

Similarly, the total number of dead females in 2000 = 1000 - (210 + 175 + 150 + 120) = 1000 - 655 = 345

Thus, the required ratio = 355 : 345 = 71 : 69.

Thus, the correct option is A.

VIDEO SOLUTION

26. **C**

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males in the age group of 30-40(in 1980) alive till 2010 = 90

The total number of females in the age group of 30-40(in 1980) alive till 2010 = 100

Thus, the total number of people in the age group of 30-40(in 1980) alive till 2010 = 90 + 100 = 190

Thus, the correct option is C.

VIDEO SOLUTION

27. **C**

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males less than 30 years (in 1980) survived until 2020 = 140 + 125 = 265

The total number of females less than 30 years (in 1980) survived until 2020 = 100 + 105 = 205

Thus, The total number of people less than 30 years (in 1980) survived until 2020 = 205 + 265 = 470

Thus, the correct option is C.

VIDEO SOLUTION

28. **40**

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males between 20 and 30 years of age in 1980 who died in 2000 = 205

The total number of males between 20 and 30 years of age in 1980 who died in 2010 = 165

Thus, the total number of males between 20 and 30 years of age in 1980 who died in the period 2000 to 2010 = 205 - 165 = 40

VIDEO SOLUTION

29. **30**
The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

We are given that there are 250 females from age 20-30 in 1980 and in 2000 these females age are from 40-50 but only 175 are alive in 2000.

In 2000 there were 175 females from age 40-50. If we assume that out of these, 30 females were of age 48 years in 2000 and they died in 2005, then there are 30 females who died at the age of 53.

If we assume that out of the 175 females, 30 females were of age 42 years in 2000, and they died in 2005, then 30 females died at the age of 47. Now, if we assume that there are 15 females of age 42 and 15 females of age 48 in the year 2000, and they all died in 2005, then we have 15 females who died at the age of 47 and 15 females who died at the age of 53.

So we can see that there are many cases possible. We are given that there were 250 females aged 20-30 in 1980, and in 2010, these females ages are from 50-60, but only 145 are alive in 2010.

In 2010 there were 145 females from age 50-60. If we assume that out of these, 40 females were of age 58 years in 2010 and they died in 2015, then there are 40 females who died at the age of 63.

If we assume that out of the 145 females, 40 females are of age 52 years age in 2010, and they died in 2015, then 40 females died at the age of 57. Now, if we assume that there are 15 females of age 52 and 25 females of age 58 in the year 2010, and they all died in 2015, then we have 15 females who died at the age of 57 and 25 females who died at the age of 63.
So we can see that again, there are many cases possible. In the first case, the range of values possible is from 0 to 30. In the second case, the range of values possible is from 0 to 40. So in total, we get a range of possible values from 0 to 70.

Thus, only one possible value of this question is not possible.

VIDEO SOLUTION

Explanation [30 - 34]:
From statement 6, it can be concluded that the total number of new cases is equal to 12+12+5+14 = 43.

Also, since the total number of cases in Kitmisto is 14, it can be concluded that the number of cases each day is either 2 or 3, where 3 cases are recorded on 4 days and 2 cases are recorded on 1 day.

From statement 4, it can be concluded that the number of new cases for Pesmisto will be 0,1,1,1, and 2, in any order(Since the total number of cases is 5, and the maximum number of new cases is 2).

In statement 3, it is given that the number of new cases kept increasing during the 5-day period.

Now, as it is already known that the maximum number of cases for Pesmisto is 2, the maximum total number of cases in a day(or on day 5) will be less than 12.

Let us consider the maximum number of cases on Day 5 as 10.

Thus the maximum number of cases possible for the remaining days will be 9, 8, 7, and 6. So, the total number of maximum cases possible for this case will be 40(less than 43).

Thus, the number of cases on Day 5 will be 11(i.e., b/w 10 and 11)

Now, if the number of cases on Day 4 is 9, the maximum number of cases possible for the remaining days will be 8, 7, and 6.

Thus, the maximum number of cases, in this case, will be 41(less than 43).

So, the number of cases on day 4 will be 10.

Now, if the number of cases on Day 3 is 8, the number of cases on day 2 will be 7, and the maximum possible number of cases on Day 1 will be 6.

Thus, the number of cases, in this case, will be 42(less than 43).

Thus, the number of cases on day 3 will be 9, the number of cases on day 2 will be 8, and the number of cases on day 1 will be 5.

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto						12
Tyhrmisto						12
Pesmisto						5
Kitmisto						14
Total	5	8	9	10	11	43

Since all the neighbourhood has at least one case on Day 1, the only possible combination will be 1, 1, 1, and 2 for Levmisto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, for the other 4 days, the number of cases in Kitmisto will be 3.

For day 5, the number of cases will be 3, 3, 2, and 3 for Levmisto, Tyhrmisto, Pesmisto and Kitmisto, respectively(since the maximum number of cases in Pesmisto is 2).

And since Pesmisto only has the maximum number of cases on one day, the number of cases on day 4 will be 3, 3, 1, and 3 for Levmisto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

On day 2, since Kitmisto is the only neighbourhood to have 3 cases, the number of cases on day 2 will be 2, 2, 1, and 3 for Levmisto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Now, on day 3, the number of cases will be 3, 3, 0, and 3 for Levmisto, Tyhrmisto, Pesmisto and Kitmisto, respectively.

Thus, the final table will be as follows:

Neighbourhoods	Day 1	Day 2	Day 3	Day 4	Day 5	Total
Levmisto	1	2	3	3	3	12
Tyhrmisto	1	2	3	3	3	12
Pesmisto	1	1	0	1	2	5
Kitmisto	2	3	3	3	3	14
Total	5	8	9	10	11	43

30. **D**
From the data, it can be concluded that the total number of cases on Day 2 is equal to 8.

Thus, the correct option is D.

VIDEO SOLUTION

31. **C**
From the final table, it can be concluded that the total number of cases in Levmisto is 3 on day 3.

Thus, the correct option is C.

VIDEO SOLUTION

32. **A**
From the final table, it can be concluded that on Day 3, the number of cases will be zero for Pesmisto.

Thus, the correct option is A.

VIDEO SOLUTION

33. **D**
From the final table, it can be concluded that both statements are false.

Thus the correct option is D.

VIDEO SOLUTION

34. **D**
It can be concluded from the final table that the number of cases will be the same for all the days.

Thus, the correct option is D

VIDEO SOLUTION

Explanation [35 - 39]:
Let the number of students in non-CS be 7x so no.of students taking AI in non-CS is 2x and no. of students taking ML in non-CS is 5x

All the CS students have taken both the courses, we are given this in the first line of the set.

From statement 5, we can conclude that 0 students from CS failed in AI and 0 students from non-CS got grade A in AI.

From statement 6, let us say that the numbers of CS students who got A, B and C grades respectively in AI were in the ratio 3a, 5a and 2a, while in ML the ratio was 4b, 5b and 2b.

From statement 3 we can say that the number of non- CS students who failed in AI and ML are b in each category.

Now from statement 8, we can say that number of CS students who failed in MI is equal to 30-b.

		A	B	C	F	Total
CS	AI	3a	5a	2a	0	10a
	ML	4b	5b	2b	30-b	10b+30
Non CS	AI	0			b	2x
	ML				b	5x

From statement 7, we can say that, $\frac{2b}{30-b} = \frac{3}{1}$

or, 2b = 90-3b

or, b= 18

CS students take both the AI and ML courses, therefore 10a= 10b +30

or, a= b+3 = 21

From statement 2, we can say that 10a = 7x

or, x = 30

Substituting the values of a, b and x in the above table.

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
Non CS	AI	0			18	60
	ML				18	150

A total of 270 students took AI out of which 252 students passed and a total of 360 students took ML out of which 330 students passed.

From statement 4, we can say that out of the 252 students who passed in AI, 126 of them got Grade B and 63 got Grade A and 63 got Grade C.

Similarly, We can say that out of the 330 students who passed in ML, 165 of them got Grade C and 99 got Grade A and 66 got Grade C.

Therefore, the final table which we get is

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
Non CS	AI	0	21	21	18	60
	ML	27	75	30	18	150

35. **D**

VIDEO SOLUTION

Explanation [40 - 44]:

Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4				
Round 5	10	10	7	13
Round 6				
Round 7		12	4	
Round 8	13			10

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11		
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3				
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3		11		10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13			10

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7		12	4	
Round 8	13	11	6	10

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6		11		
Round 7	12	12	4	12
Round 8	13	11	6	10

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

	Pulak	Qasim	Ritesh	Suresh
Round 1	12	8	10	10
Round 2	13	10	9	8
Round 3	11	11	8	10
Round 4	11	11	6	12
Round 5	10	10	7	13
Round 6	12	11	5	12
Round 7	12	12	4	12
Round 8	13	11	6	10

40. **D**

VIDEO SOLUTION

42. **6**

VIDEO SOLUTION

43. **6**

The Games which had a bet of Rs.2 are as following
Round 1 - Palak vs Qasim
Round 2 - Qasim vs Suresh
Round 3 - Palak vs Suresh
Round 4 - Ritesh vs Suresh
Round 5 - None
Round 6 - Palak vs Ritesh
Round 7 - None
Round 8 - Ritesh vs Suresh
Hence total number of games that were played with a bet of 2 is 6

VIDEO SOLUTION

44. **D**

For round 6 the pairs formed were Pulak-Ritesh and Qasim-Suresh. For round 4 the pairs formed were Pulak-Qasim and Ritesh-Suresh.

Therefore the pairs formed for round 5 were Pulak-Suresh and Qasim-Ritesh

VIDEO SOLUTION

Quant

45. **B**

Let $\frac{x}{y}$ $be\ t$

Therefore, $c = 16t\ +\ \frac{49}{t}$

Applying AM>= GM

$\frac{\left(16t\ +\ \frac{49}{t}\right)}{2} \geq\ \left(16t\ \times\ \frac{49}{t}\right)^{\frac{1}{2}}$

$16t\ +\ \frac{49}{t} \geq 56$

When t is positive then c is greater than equal to 56.

When t is negative then c is less than equal to -56.

Therefore $c\ \in\ \left(-\infty,\ -56\right]\ \cup\ \left[56,\infty\ \right]$

As -50 is not in the range of c so it is the answer

VIDEO SOLUTION

46. **A**

$2x^2 + kx + 5 = 0$ has no real roots so D<0

$k^2 - 40\ \ < 0$

$\left(k - \sqrt{40}\right)\left(k + \sqrt{40}\right) < 0$

$k \in \left(-\sqrt{40}, \sqrt{40}\right)$

$x^2 + (k-5)x + 1 = 0$ has two distinct real roots so D>0

$\left(k-5\right)^2 - 4 > 0$

$k^2 - 10k + 21 > 0$

$(k-3)(k-7) > 0$

$k \in (-\infty\ , 3) \cup (7, \infty\ )$

Therefore possibe value of k are -6, -5, -4, -3, -2, -1, 0, 1, 2

In 9 total 9 integer values of k are possible.

VIDEO SOLUTION

47. **14**

$(\sqrt{\frac{7}{5}})^{3x-y} = \frac{875}{2401}$

$(\frac{7}{5})^{\frac{(3x-y)}{2}} = \frac{125}{343}$

$(\frac{7}{5})^{\frac{(3x-y)}{2}} = (\frac{7}{5})^{-3}$

3x-y = -6

$(\frac{4a}{b})^{6x-y} = (\frac{2a}{b})^{y-6x}$

Therefor, y=6x as the bases are different so the power should be zero for the results to be equal.

3x-y=-6

or, 3x - 6x = -6

or x= 2

y= 6x = 12

x+y = 14

VIDEO SOLUTION

48. **D**

Let the six numbers be a, b, c, d, e, f in ascending order

a+b = 28

e+f = 56

If we want to maximise the average then we have to maximise the value of c and d and maximise e and minimise f

e+f = 56

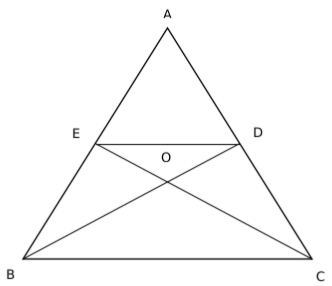
As e and f are distinct natural numbers so possible values are 27 and 29

Therefore c and d will be 25 and 26 respecitively

So average = $\frac{(a+b+c+d+e+f)}{6} = \frac{(28+25+26+56)}{6} = \frac{135}{6} = 22.5$

VIDEO SOLUTION

49. **9**



Area of ABD : Area of BDC = 1:1

Therefore, area of ABD = 54

Area of ADE : Area of EDB = 1:1

Therefore, area of ADE = 27

O is the centroid and it divides the medians in the ratio of 2:1

Area of BEO : Area of EOD = 2:1

Area of EOD = 9

VIDEO SOLUTION

50. **D**

a, b, c, m and n are integers so if one root is $3 + 2\sqrt{2}$ then the other root is $3 - 2\sqrt{2}$

Sum of roots = 6 = -b/a or b= -6a

Product of roots = 1 = c/a or c=a

a, b, c, m and n are integers so if one root is $4 + 2\sqrt{3}$ then the other root is $4 - 2\sqrt{3}$

Sum of roots = 8 = -m/a or m = -8a

product of roots = 4 = n/a or n = 4a

$(\frac{b}{m}+\frac{c-2b}{n})$

= $\frac{6a}{8a}+\frac{(a+12a)}{4a}=\frac{3}{4}+\frac{13}{4}=\frac{16}{4}=4$

VIDEO SOLUTION

51. **D**

Let the unit of work done by 1 man in 1 hour and 1 day be 1 MDH unit (Man Day Hour).

Thus, in 7 hours per day for 10 days, the work done by N people =$N\times\ 7\times\ 10$ MDH units.

Since this is equal to 35% of the total work,

35% of the total work = $N\times\ 7\times\ 10$ MDH units.

Total work = $\frac{(N\times\ 100\times\ 7\times\ 10)}{35}=200\times\ N$ MDH units.

The work left = $200N-70N=130N$ MDH units.

Now, 10 people left the job. So, the number of people left = (N-10)

Since (N-10) people completed the rest of work in 14 days by working 10 hours a day,

$(N-10)\times\ 14\times\ 10=130N$

$10N=1400$

N = 140

Thus, the correct option is D.

VIDEO SOLUTION

52. **A**

Initially: a glass 500cc milk and a cup 500cc water

Step 1: 150 cc of milk is transferred to the cup from glass

After step 1: Glass - 350 cc milk, Cup - 150 cc milk and 500 cc water

Step 2: 150 cc of this mixture is transferred from the cup to the glass

After step 2:

Glass - 350 cc milk + 150 cc mixture with milk:water ratio 3:10

Cup - 500 cc mixture with milk:water ratio 3:10

water in glass : milk in cup = $\frac{10}{13}\times 150\ :\ \frac{3}{13}\times 500=1:1$

The answer is option A.

VIDEO SOLUTION

53. **B**

It is given,

$\frac{5.5\times1\times8000}{100}+\frac{5.6\times1\times5000}{100}+\frac{x\times1\times7000}{100}=\frac{5}{100}\times 20000$

$440+280+70x=1000$

x = 4%

Interest = $\frac{20000\times4\times1}{100}$ = Rs 800

The answer is option B.

VIDEO SOLUTION

54. **B**

Both the cars take 1.5 hrs to meet when they travel towards each other.

It is given, speed of slower car is 60 km/hr

Therefore, distance covered by slower car before they meet = 60*1.5 = 90 km

The answer is option B.

VIDEO SOLUTION

55. **A**

The number of 4-digit numbers possible using 1,1,2, and 4 is $\frac{4!}{2!}=12$

Number of 1's, 2's and 4's in units digits will be in the ratio 2:1:1, i.e. 6 1's, 3 2's and 3 4's.

Sum = 6(1) + 3(2) + 3(4) = 24

Similarly, in tens digit, hundreds digit and thousands digit as well.

Therefore, sum = 24 + 24(10) + 24(100) + 24(1000) = 24(1111)

Mean = $\frac{24(1111)}{12} = 2222$

The answer is option A.

VIDEO SOLUTION

56. **D**

Sum of the three sides of a quadrilateral is greater than the fourth side.

Therefore, let the fourth side be

1+2+4>d or d<7

1+2+d>4 or d>1

Possible values of d are 2, 3, 4, 5 and 6.

VIDEO SOLUTION

57. **548**

General term = 38 + (n-1)17 = 17n + 21 = 17(n+1) + 4 = 17k + 4

Each term is in the form of 17k + 4

Least 3-digit number in the form of 17k + 4 is at k = 6, i.e. 106

Highest 3-digit number in the form of 17k + 4 is at k = 58, i.e. 990

106, 123, 140,..........., 990

990 = 106 + 17(n-1)

n = 53

Sum = $\frac{53}{2}(106 + 990) = 53 \times 548$

Average = $53 \times \frac{548}{53} = 548$

VIDEO SOLUTION

58. **63**

Let the number of students in section A and B be a and b, respectively.

It is given, a = b - 10

$\frac{32a + 60b}{a + b}$ is an integer

$$\frac{32a + 60(a+10)}{a + a + 10} = k$$

$$\frac{46a + 300}{a + 5} = k$$

$$k = \frac{46(a+5)}{a+5} + \frac{70}{a+5}$$

$$k = 46 + \frac{70}{a+5}$$

a can take values 2, 5, 9, 30, 65

Difference = 65 - 2 = 63

VIDEO SOLUTION

59. **B**

When x< r

f(x) = r

f(x) = f(f(x))

r = f(r)

r= 2r-r

r=r

When x>=r

f(x) = 2x-r

f(x) = f(f(x))

2x-r = f(2x-r)

2x-r = 2(2x-r) - r

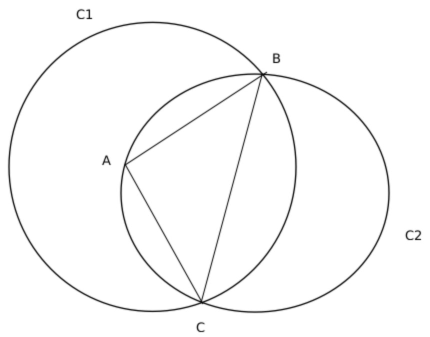
2x-r = 4x-3r

or, x=r

Therefore x<= r

VIDEO SOLUTION

60. **C**



BC is the diameter of circle C2 so we can say that $\angle BAC = 90^\circ$ as angle in the semi circle is 90°

Therefore overlapping area = $\frac{1}{2}$(Area of circle C2) + Area of the minor sector made be BC in C1

AB= AC = 8 cm and as $\angle BAC = 90^\circ$, so we can conclude that BC= $8\sqrt{2}$ cm

Radius of C2 = Half of length of BC = $4\sqrt{2}$ cm

Area of C2 = $\pi\left(4\sqrt{2}\right)^2 = 32\pi\ cm^2$

A is the centre of C1 and C1 passes through B, so AB is the radius of C1 and is equal to 8 cm

Area of the minor sector made be BC in C1 = $\frac{1}{4}$(Area of circle C1) - Area of triangle ABC = $\frac{1}{4}\pi\left(8\right)^2 - \left(\frac{1}{2} \times 8 \times 8\right) = 16\pi - 32\ cm^2$

Therefore,

Overlapping area between the two circles= $\frac{1}{2}$(Area of circle C2) + Area of the minor sector made be BC in C1

= $\frac{1}{2}\left(32\pi\right) + \left(16\pi - 32\right) = 32\left(\pi - 1\right) cm^2$

VIDEO SOLUTION

61. **150**

Since the total number of students, when divided by either 9 or 10 or 12 or 25 each, gives a remainder of 4, the number will be in the form of LCM(9,10,12,25)k + 4 = 900k + 4.

It is given that the value of 900k + 4 is less than 5000.

Also, it is given that 900k + 4 is divided by 11.

It is only possible when k = 2 and total students = 1804.

So, the number of 12 students group = 1800/12 = 150.

VIDEO SOLUTION

62. **B**

Let $\frac{x^2-6x+10}{3-x} = p$

$x^2 - 6x + 10 = 3p - px$

$x^2 - \left(6 - p\right)x + 10 - 3p = 0$

Since the equation will have real roots,

$\left(6 - p\right)^2 - 4 \times \left(10 - 3p\right) \geq 0$

$p^2 - 12p + 12p + 36 - 40 \geq 0$

$p^2 \geq 4$

$p \geq 2$, $p \leq -2$

Now, when $p = -2, x = 4$. Since it is given that $x < 3$, thus this value will be discarded.

Now, $\frac{1}{2}$ and $-\frac{1}{2}$ do not come in the mentioned range.

when $p = 2, x = 2$

Thus, the minimum possible value of p will be 2.

Thus, the correct option is B.

Alternate explanation:

Since $x < 3$,

$3 - x > 0$

Let $3-x=y$. So, $y>0$.

Now, $\frac{x^2-6x+10}{3-x} = \frac{x^2-6x+9+1}{3-x}$

=> $\frac{(3-x)^2+1}{3-x}$

Since $3-x=y$, the equation will transform to $\frac{y^2+1}{y}$ or $y+\frac{1}{y}$

The minimum value of the expression $y+\frac{1}{y}$ for $y>0$ will at $y=1$

i.e., **Minimum value** = $1+1=2$

Thus, the correct option is B.

VIDEO SOLUTION

63. **12**

Let the number of 100 cheques, 250 cheques and 500 cheques be x, y and z respectively.

We need to find the maximum value of z.

x + y + z = 100 ...... (1)

100x + 250y + 500z = 15250

2x + 5y + 10z = 305 ...... (2)

2x + 2y + 2z = 200 .... (1)

(2) - (1), we get

3y + 8z = 105

At z = 12, x = 3

Therefore, maximum value z can take is 12.

VIDEO SOLUTION

64. **60**

Let the speed of Moody be 'x' steps/sec and that of the escalator be 'y' steps/sec.

In 30 seconds, Moody will finish riding the escalator when going in the same direction.

Thus, total steps = 30(x+y)

If Moody's speed becomes twice, the time becomes 20 seconds.

Thus, total steps = 20(2x+y)

Or 30x + 30y = 40x + 20y

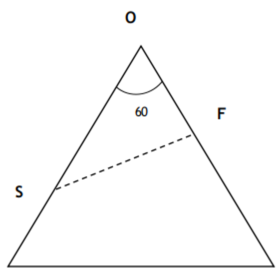
Or x = y

So, total steps = 60y.

Time taken by only escalator= 60y/y = 60s.

VIDEO SOLUTION

65. **B**



Let S be the slower ship and F be the faster ship.

It is given that when S travelled 8 km, the positions of ships with the port is forming a right triangle.

Since one of the angles is 60(since one vertex is still part of the equilateral triangle),

the other two vertexes will have angles of 30 and 90.

The distance between O and S = 24 - 8 = 16

In triangle OFS, $\cos 60^0 = \frac{OF}{OS}$

Thus, OF = 8.

Thus in the time, S covered 8 km, F will cover 24 - 8 = 16 km.

Thus, the ratio of their speeds is 2:1,

Thus, when F covers 24 km, S will cover 12 km.

The correct option is B.

VIDEO SOLUTION

66. **11**

Let the efficiency of Bob be 3 units/day. So, Alex's efficiency will be 6 units/day, and Cole's will be 2 units/day.

Since Bob can finish the job in 40 days, the total work will be 40*3 = 120 units.

Since Alex and Bob work on the first day, the total work done = 3 + 6 = 9 units.

Similarly, for days 2 and 3, it will be 5 and 8 units, respectively.

Thus, in the first 3 days, the total work done = 9 + 5 + 8 = 22 units.

The work done in the first 15 days = 22*5 = 110 units.

Thus, the work will be finished on the 17th day(since 9 + 5 = 14 units are greater than the remaining work).

Since Alex works on two days of every 3 days, he will work for 10 days out of the first 15 days.

Then he will also work on the 16th day.

The total number of days = 11.

VIDEO SOLUTION